Description

Image



Caption

PTFE is widely used for non-stick pans and spatulas. © ANSYS, Inc.

The material

PTFE (Teflon) is a member of the fluoroplastic family, which includes chlorotrifluoroethylene, CTFE or CFE, polyvinyl fluoride, PVF, and polyvinylidene fluoride PVF2. PTFE has exceptionally low friction, is water repellant, and extremely stable. It was first commercialized in the late 1940's as Teflon. Non-stick cooking utensils (Tefal = Teflon coated aluminum) exploiting its chemical inertness, its thermal stability and its non-wettability - the reason nothing sticks to it. It is expensive as polymers go, but it is used in high-value applications (non-stick pans; Gore-Tex rain gear; artificial arteries).

Compositional summary

(CF2-CF2)n

General properties

Density	2.14e3	-	2.2e3	kg/m^3
Price	* 8.08	-	13	GBP/kg
Date first used	1938			

Mechanical properties

Young's modulus	0.4	-	0.552	GPa
Shear modulus	* 0.138	-	0.19	GPa
Bulk modulus	* 1.53	-	1.6	GPa
Poisson's ratio	0.44	-	0.46	
Yield strength (elastic limit)	19.7	-	21.7	MPa
Tensile strength	20.7	-	34.5	MPa
Compressive strength	* 11.2	-	12.3	MPa

Elongation		200	-	400	% strain
Hardness - Vickers	*	6	-	7	HV
Fatigue strength at 10^7 cycles		5.75	-	7	MPa
Fracture toughness	*	1.32	-	1.8	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0725	-	0.1	

Thermal properties

Melting point		315	-	339	°C
Glass temperature		117	-	130	°C
Maximum service temperature		250	-	271	°C
Minimum service temperature		-268	-	-200	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.242	-	0.261	W/m.°C
Specific heat capacity		970	-	1.09e3	J/kg.°C
Thermal expansion coefficient		120	-	170	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		3.3e23	-	3e24	µohm.cm
Dielectric constant (relative permittivity)		2	-	2.2	
Dissipation factor (dielectric loss tangent)		2e-4	-	3e-4	
Dielectric strength (dielectric breakdown)		18.2	-	19.7	MV/m

Optical properties

Transparency		Translucent			
Refractive index		1.31	-	1.36	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4			
Machinability		3	-	4	
Weldability		3	-	4	

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			

Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent

Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Excellent
O2 (oxygen gas)	Acceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Good

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 15.4	-	17	kg/kg
Embodied energy (virgin grade production)	* 195	-	215	MJ/kg
Water usage	* 434	-	480	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.24	-	9.18	MJ/kg
Polymer molding (injection, blow), embodied energy	* 26.2	-	31.3	MJ/kg
Machining, coarse, embodied energy	* 14.9	-	16.5	MJ/kg
Machining, fine, embodied energy	* 15.5	-	17.2	MJ/kg

Machining, grinding, embodied energy	* 16.1	-	17.9	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.545	-	0.61	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.88	-	2.28	kg/kg
Machining, coarse, climate change (CO2-eq)	* 0.975	-	1.08	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.01	-	1.13	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.05	-	1.17	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	0.672	-	0.742	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

PTFE is non-flammable and FDA approved. Like all thermoplastics, simple PTFE can be recycled. But in making it into non-stick surfaces, or in transforming it into Gore-Tex, additives are made which prevent further recycling.

Recycle mark



Supporting information

Design guidelines

PTFE is 2.7 times denser than polyethylene and 12 times more expensive. But it is much more resistant to chemical attack; it can safely be used from -270 to + 250 C. It has remarkably low friction; and it has an exceptional ability to resist wetting. All fluoroplastics are white, and to some degree, translucent. They give long-term resistance to attacks of all sorts, including ultraviolet radiation. PTFE itself has a characteristically soft, waxy feel, partly because of the low coefficient of friction. It is an excellent electrical insulator, with low dielectric loss. It can be "foamed" to give a light, micro-porous film that rejects liquid water but allows water vapor to pass - the principle of Gore-Tex. The mechanical properties of PTFE are not remarkable, but it can be made more abrasive resistant by filling with inert ceramic and it can be reinforced with glass, nylon or Kevlar fibers to give a leather-like skin of exceptional toughness, strength and weather-resistance (exploited in tensile roofs). Bonding PTFE is difficult; thermal or ultrasonic methods are good; epoxy, nitrile-phenolic and silicone adhesives can be used. The use of Gore-Tex derivatives in fabrics is expanding, with new variants being developed. The pore size in these fabrics can be controlled to reject not merely water, but bacteria, with potential for protective clothing for surgeons, and possibly against certain kinds of biological weapons. PTFE itself has FDA approval. Its architectural use for dramatic, tent-like, roofing of large structures is increasing.

Technical notes

Fluorine is the most reactive of gasses, yet combined with carbon to form fluoropolymers and it becomes the most stable of molecules, resistant to practically everything except excessive heat. Polytetrafluoroethylene, PTFE, the simplest of these, is just polyethylene with all the hydrogens stripped off and replaced by fluorine: (-CF2)n. The others are variants on this.

Typical uses

Wire and cable covers, high-quality insulating tape, corrosion resistant lining for pipes and valves, protective coatings, seals and gaskets, low friction bearings and skis, transparent roofing and weather protection for other polymers (e.g. ABS), non-stick cooking products, water repellent fabrics.

Tradenames

Aflas, Algoflon, Duroid, Dyneon, Fluon, Fluorel, Hostaflon TF, Polyflon, Soreflon, THV, Teflon, Tetraflon, Tetrafluor

Links

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Reference